

STANDARD 2

Stage 6

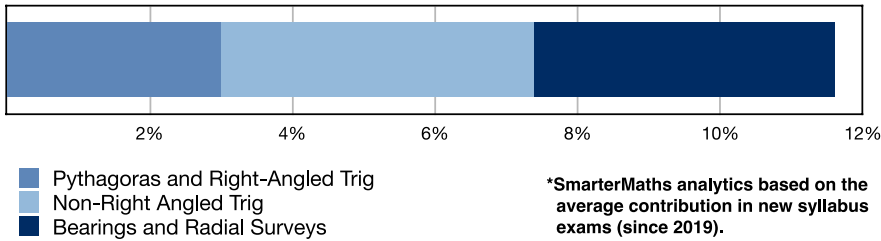
Measurement (Std 2), M6 Trigonometry (Y12)

Bearings and Radial Surveys (Std2)

Teacher: Hailey Summer

Exam Equivalent Time: 129 minutes (based on allocation of 1.5 minutes per mark)

STD2 Exam Contribution History M6 - Trigonometry



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- MS-M6 Trigonometry* has been a very significant contributor to new syllabus Std2 exams, responsible for an average of 11.6% per exam since 2019.
- We have split this area into 3 subtopics: *Pythagoras and Right-Angled Trig* (3.0%), *Non-Right-Angled Trig* (4.4%) and *Bearings and Radial Surveys* (4.2%).
- This analysis looks at the sub-topic, *Bearings and Radial Surveys*.

ANALYSIS - What to Expect and Common pitfalls

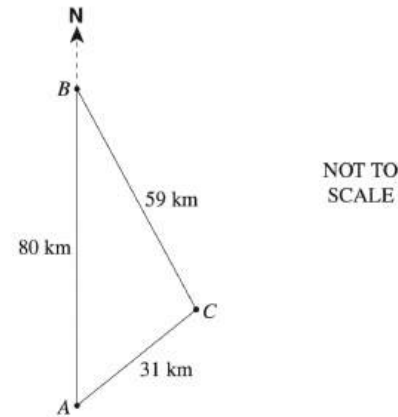
- Bearings and Radial Surveys* (4.2%) have attracted healthy allocations of at least 4 marks in new syllabus Std2 exams to date.
- Bearings* questions have been the most common question type, appearing three times in the last four years. On each occasion, sub-50% state mean marks have resulted, making this a key revision focus area.
- True Bearings* is an important concept to understand well and has been examined three times since 2019 (see 2022 8 MC, 2021 4 MC and 2019 4 MC)
- Radial Surveys* have been most recently examined in 2021 and 2019. On both occasions, students have shown a general lack of understanding. Carefully review these questions.
- HSC Markers' Comment of note: More successful students draw North/South parallel lines through important points on *large clear diagrams*. Simple advice that is incorporated into worked solutions. Follow it.

Questions

1. Measurement, STD2 M6 2012 HSC 20 MC

Town **B** is 80 km due north of Town **A** and 59 km from Town **C**.

Town **A** is 31 km from Town **C**.

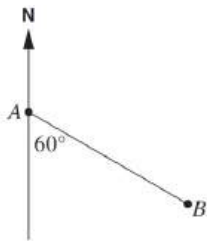


What is the bearing of Town **C** from Town **B**?

- A. 019°
- B. 122°
- C. 161°
- D. 341°

2. Measurement, STD2 M6 2006 HSC 13 MC

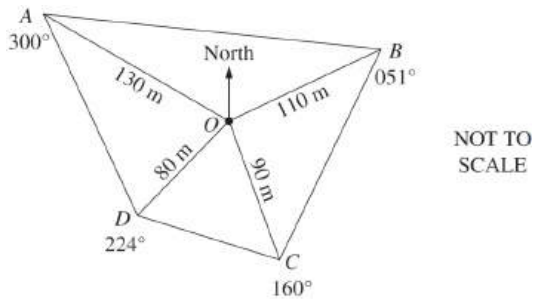
What is the bearing of **A** from **B**?



- A. 060°
- B. 120°
- C. 150°
- D. 300°

3. Measurement, STD2 M6 2015 HSC 7 MC

The diagram shows a radial survey of a field **ABCD**.



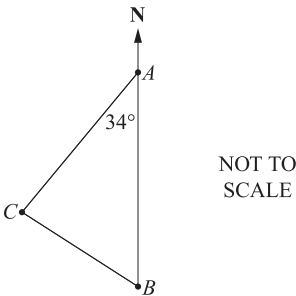
In triangle **AOB**, what is the size of $\angle AOB$?

- A. 51°
- B. 111°
- C. 125°
- D. 249°

4. Measurement, STD2 M6 2018 HSC 7 MC

The diagram shows the positions of towns **A**, **B** and **C**.

Town **A** is due north of town **B** and $\angle CAB = 34^\circ$



What is the bearing of town **C** from town **A**?

- A. 034°
- B. 146°
- C. 214°
- D. 326°

5. Measurement, STD2 M6 2022 HSC 8 MC

Which true bearing is the same as **S48°W**?

- A. 132°
- B. 222°
- C. 228°
- D. 312°

6. Measurement, STD2 M6 SM-Bank 2 MC

Which of the following expresses S25°E as a true bearing?

- A. 025°
- B. 065°
- C. 115°
- D. 155°

7. Measurement, STD2 M6 SM-Bank 3 MC

Which of the following expresses S65°W as a true bearing?

- A. 065°
- B. 155°
- C. 245°
- D. 295°

8. Measurement, STD2 M6 SM-Bank 7 MC

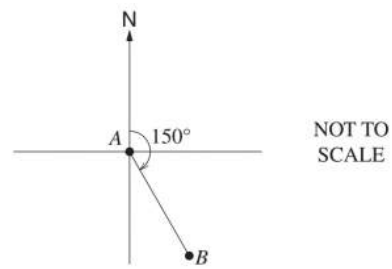
Jeet walks 5 km from his home on a bearing of 153°. He then walks due north until he arrives a point which is due east of his home.

How far east, to the nearest 0.1 km, is Jeet from home?

- A. 2.3 km
- B. 2.5 km
- C. 4.9 km
- D. 9.8 km

9. Measurement, STD2 M6 2010 HSC 10 MC

A plane flies on a bearing of 150° from **A** to **B**.

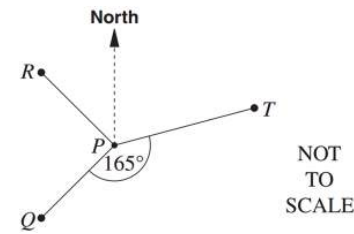


What is the bearing of **A** from **B**?

- A. 30°
- B. 150°
- C. 210°
- D. 330°

10. Measurement, STD2 M6 2008 HSC 17 MC

The diagram shows the position of **Q**, **R** and **T** relative to **P**.



In the diagram,

Q is south-west of **P**

R is north-west of **P**

$\angle QPT$ is 165°

What is the bearing of **T** from **P**?

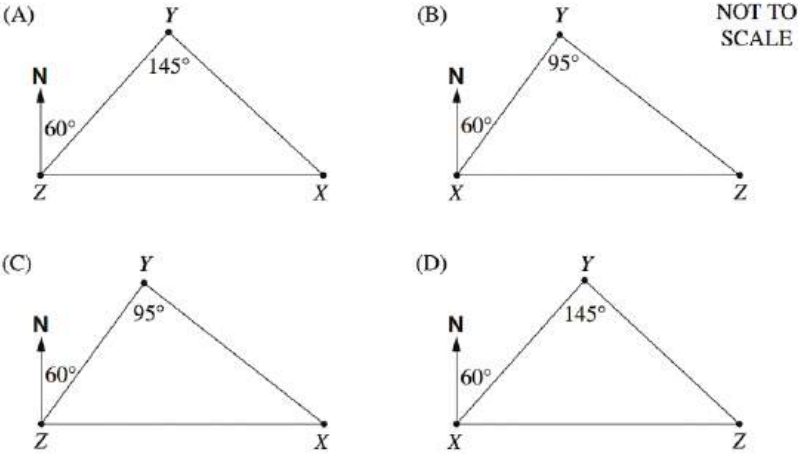
- A. 060°
- B. 075°
- C. 105°
- D. 120°

11. Measurement, STD2 M6 2014 HSC 23 MC

The following information is given about the locations of three towns ***X***, ***Y*** and ***Z***:

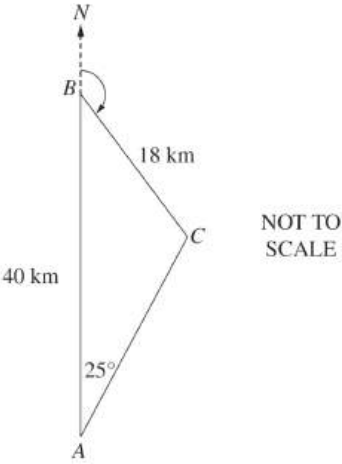
- ***X*** is due east of ***Z***
- ***X*** is on a bearing of 145° from ***Y***
- ***Y*** is on a bearing of 060° from ***Z***.

Which diagram best represents this information?



12. Measurement, STD2 M6 2016 HSC 25 MC

The diagram shows towns ***A***, ***B*** and ***C***. Town ***B*** is 40 km due north of town ***A***. The distance from ***B*** to ***C*** is 18 km and the bearing of ***C*** from ***A*** is 025° . It is known that $\angle BCA$ is obtuse.



What is the bearing of ***C*** from ***B***?

- A. 070°
- B. 095°
- C. 110°
- D. 135°

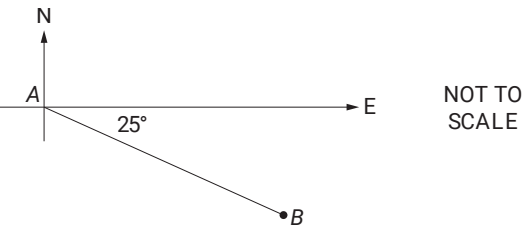
13. Measurement, STD2 M6 2019 HSC 4 MC

Which compass bearing is the same as a true bearing of 110° ?

- A. $S20^\circ E$
- B. $S20^\circ W$
- C. $S70^\circ E$
- D. $S70^\circ W$

14. Measurement, STD2 M6 2021 HSC 14 MC

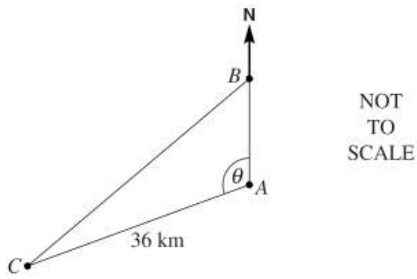
Consider the diagram below.



What is the true bearing of **A** from **B**?

- A. 025°
- B. 065°
- C. 115°
- D. 295°

15. Measurement, STD2 M6 2005 HSC 27c

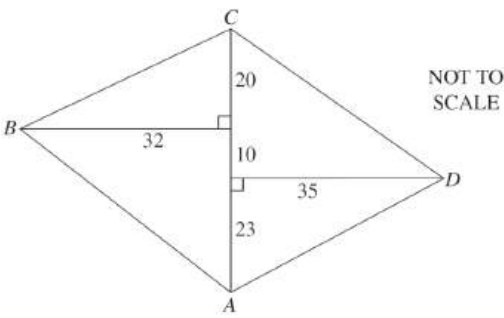


The bearing of **C** from **A** is 250° and the distance of **C** from **A** is 36 km.

- i. Explain why θ is 110°. (1 mark)
- ii. If **B** is 15 km due north of **A**, calculate the distance of **C** from **B**, correct to the nearest kilometre. (3 marks)

16. Measurement, STD2 M6 2016 HSC 26d

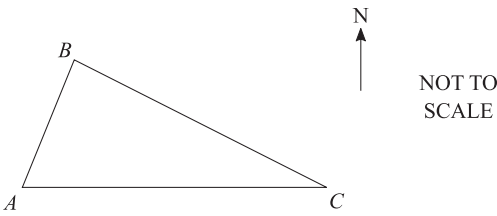
The diagram shows a block of land **ABCD** that has been surveyed. All measurements are in metres.



Calculate the length of **AB**, correct to the nearest metre. (2 marks)

17. Measurement, STD2 M6 SM-Bank 4

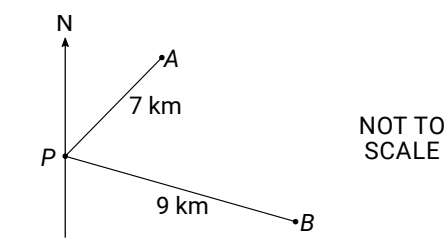
The diagram shows three checkpoints **A**, **B** and **C**. Checkpoint **C** is due east of Checkpoint **A**. The bearing of Checkpoint **B** from Checkpoint **A** is N22°E and the bearing of Checkpoint **C** from Checkpoint **B** is S68°E. The distance between Checkpoint **A** and Checkpoint **B** is 42 kilometres.



- i. Mark the given information on the diagram and explain why $\angle ABC$ is 90°. (2 marks)
- ii. Find the distance, to the nearest kilometre, between Checkpoint **A** and Checkpoint **C**. (2 marks)
- iii. If a runner is travelling 12.6 km/h, how long does it take her to travel between Checkpoint **A** and Checkpoint **B**, in hours and minutes? (2 marks)

18. Measurement, STD2 M6 2020 HSC 31

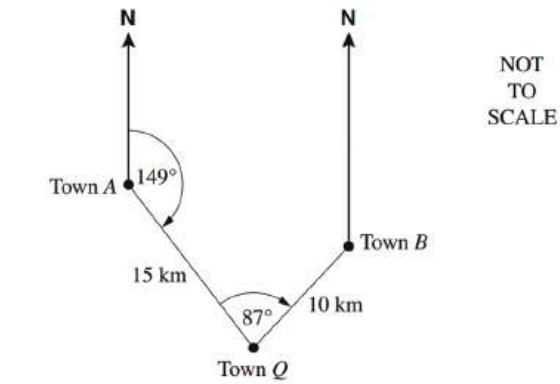
Mr Ali, Ms Brown and a group of students were camping at the site located at **P**. Mr Ali walked with some of the students on a bearing of 035° for 7 km to location **A**. Ms Brown, with the rest of the students, walked on a bearing of 100° for 9 km to location **B**.



- a. Show that the angle **APB** is 65° . (1 mark)
- b. Find the distance **AB**. (2 marks)
- c. Find the bearing of Ms Brown's group from Mr Ali's group. Give your answer correct to the nearest degree. (2 marks)

19. Measurement, STD2 M6 2007 HSC 26a

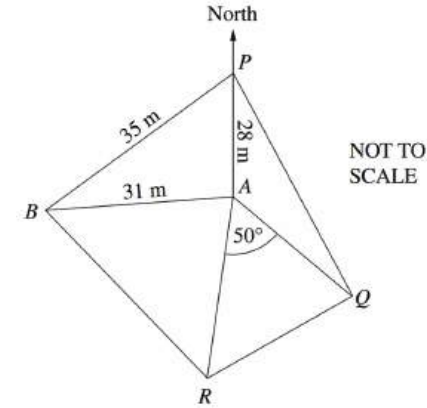
The diagram shows information about the locations of towns **A**, **B** and **Q**.



- i. It takes Elina 2 hours and 48 minutes to walk directly from Town **A** to Town **Q**. Calculate her walking speed correct to the nearest km/h. (1 mark)
- ii. Elina decides, instead, to walk to Town **B** from Town **A** and then to Town **Q**. Find the distance from Town **A** to Town **B**. Give your answer to the nearest km. (2 marks)
- iii. Calculate the bearing of Town **Q** from Town **B**. (1 mark)

20. Measurement, STD2 M6 2004 HSC 24b

The diagram shows a radial survey of a piece of land.



- i. **Q** is south-east of **A**. What is the size of angle **PAQ**? (1 mark)
- ii. What is the bearing of **R** from **A**? (1 mark)
- iii. Find the size of angle **PAB** to the nearest degree. (3 marks)

21. Measurement, STD2 M6 2006 HSC 26a

Daniel conducts an offset survey to sketch a diagram, **ABCD**, of a block of land.

Daniel walks from **A** to **C**, a distance of 62 m.

When he is 15 m from **A**, he notes that point **D** is 25 m to his right.

When he is 57 m from **A**, he notes that point **B** is 20 m to his left.

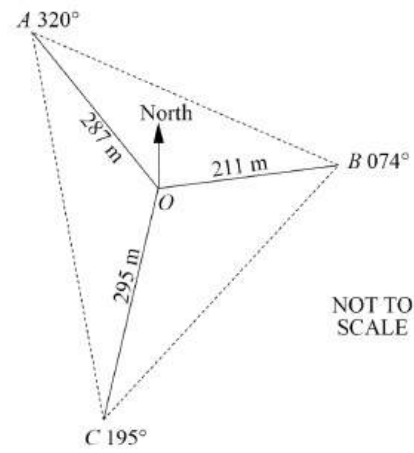
This is his notebook entry.

		C	
	62		
B 20	57		
	15	25 D	
	0		
	A		

- i. Draw a neat sketch of the block of land. Label **A**, **B**, **C** and **D** on your diagram. (1 mark)
- ii. Calculate the distance from **C** to **D**. (Give your answer to the nearest metre.) (2 marks)

22. Measurement, STD2 M6 2014 HSC 28b

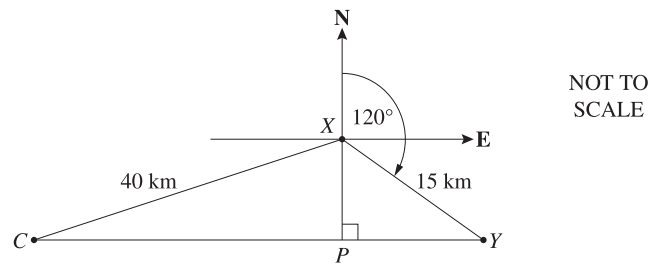
A radial compass survey of a sports centre is shown in the diagram.



- i. Show that the size of angle AOB is 114° . (1 mark)
- ii. Calculate the length of the boundary AB , to the nearest metre. (2 marks)
- iii. Find the area of triangle AOB in hectares, correct to two significant figures. (3 marks)

23. Measurement, STD2 M6 2023 HSC 27

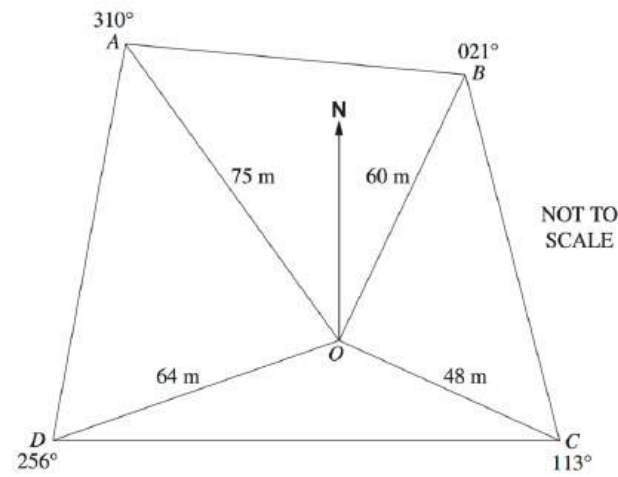
The diagram shows the location of three places X , Y and C .
 Y is on a bearing of 120° and 15 km from X .
 C is 40 km from X and lies due west of Y .
 P lies on the line joining C and Y and is due south of X .



- a. Find the distance from X to P . (2 marks)
- b. What is the bearing of C from X , to the nearest degree? (2 marks)

24. Measurement, STD2 M6 2013 HSC 28a

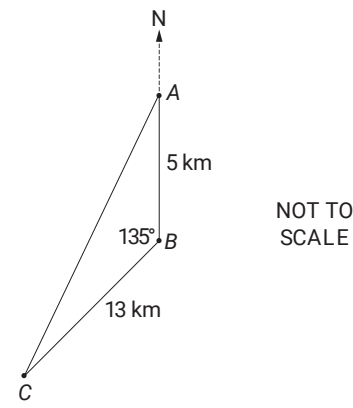
A compass radial survey of the field $ABCD$ has been conducted from O .



Find the area of the section ABO , to the nearest square metre. (2 marks)

25. Measurement, STD2 M6 2017 HSC 30c

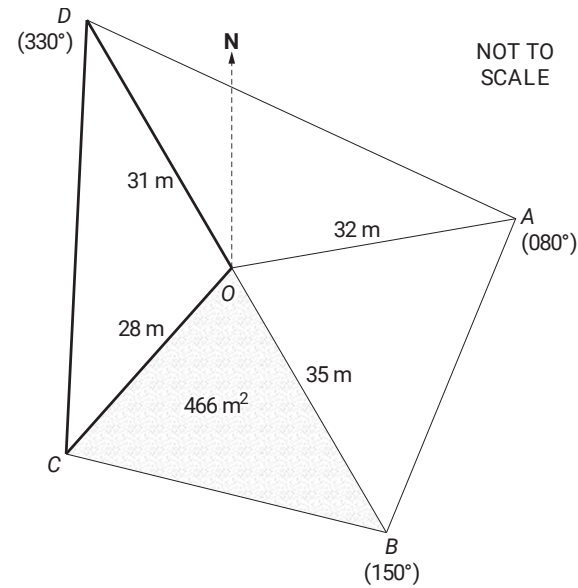
The diagram shows the location of three schools. School A is 5 km due north of school B , school C is 13 km from school B and $\angle ABC$ is 135° .



- i. Calculate the shortest distance from school A to school C , to the nearest kilometre. (2 marks)
- ii. Determine the bearing of school C from school A , to the nearest degree. (3 marks)

26. Measurement, STD2 M6 2021 HSC 39

The diagram shows a compass radial survey of the field **ABCD**.



a. Triangle **COB** has an area of 466 m².

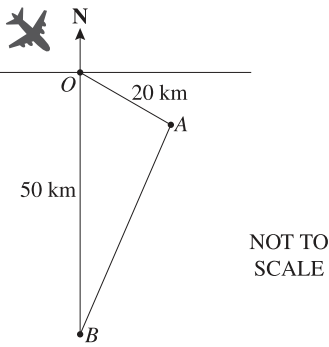
Find the size of acute angle **COB**, correct to the nearest degree. (2 marks)

b. A farmer wants to put a fence around the triangle **DOC**.

Find the length of fencing required. Give your answer in metres correct to one decimal place. (3 marks)

27. Measurement, STD2 M6 2022 HSC 33

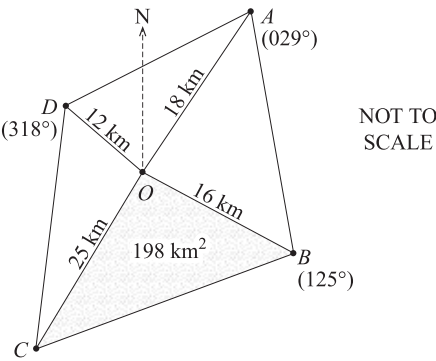
The diagram shows an aeroplane that was flying towards an airport at **A** on a bearing of **135°T**. When it was at point **O**, 20 km away from the airport at **A**, the flight course was changed. The aeroplane landed at an airport at **B** directly south of **O**. The distance from **O** to **B** is 50 km.



- a. Show that the distance between the airport at **A** and the airport at **B** is 38.5 km, correct to 1 decimal place. (2 marks)
- b. Use the sine rule to find the angle **OBA** to the nearest degree. (2 marks)
- c. What is the bearing of the airport at **B** from the airport at **A**? (1 mark)

28. Measurement, STD2 M6 2019 HSC 35

A compass radial survey shows the position of four towns **A**, **B**, **C** and **D** relative to point **O**.

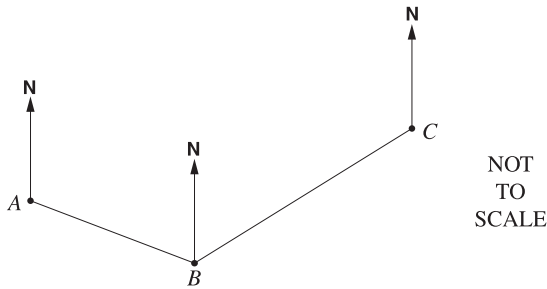


The area of triangle **BOC** is 198 km².

Calculate the bearing of town **C** from point **O**, correct to the nearest degree. (3 marks)

29. Measurement, STD2 M6 2011 HSC 24c

A ship sails 6 km from **A** to **B** on a bearing of 121° . It then sails 9 km to **C**. The size of angle **ABC** is 114° .

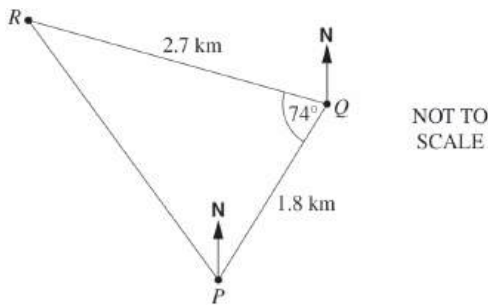


- i. What is the bearing of **C** from **B**? (1 mark)
- ii. Find the distance **AC**. Give your answer correct to the nearest kilometre. (2 marks)
- iii. What is the bearing of **A** from **C**? Give your answer correct to the nearest degree. (3 marks)

30. Measurement, STD2 M6 2009 HSC 27b

A yacht race follows the triangular course shown in the diagram. The course from **P** to **Q** is 1.8 km on a true bearing of 058° .

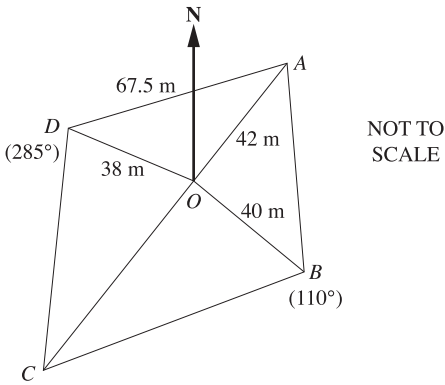
At **Q** the course changes direction. The course from **Q** to **R** is 2.7 km and $\angle PQR = 74^\circ$.



- i. What is the bearing of **R** from **Q**? (1 mark)
- ii. What is the distance from **R** to **P**? (2 marks)
- iii. The area inside this triangular course is set as a 'no-go' zone for other boats while the race is on.
What is the area of this 'no-go' zone? (1 mark)

31. Measurement, STD2 M6 2024 HSC 40

A compass radial survey is shown.



Given that **AC** is a straight line, find the bearing of **C** from **O**, correct to the nearest degree. (3 marks)

Worked Solutions

1. Measurement, STD2 M6 2012 HSC 20 MC

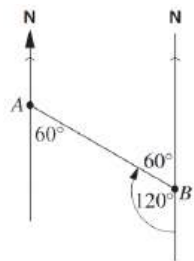
Using the cosine rule:

$$\begin{aligned}\cos \angle B &= \frac{a^2 + c^2 - b^2}{2ac} \\ &= \frac{59^2 + 80^2 - 31^2}{2 \times 59 \times 80} \\ &= 0.9449 \dots\end{aligned}$$

$$\angle B = 19^\circ \text{ (nearest degree)}$$

$$\begin{aligned}\therefore \text{Bearing of Town C from B} &= 180 - 19 = 161^\circ \\ &\Rightarrow C\end{aligned}$$

2. Measurement, STD2 M6 2006 HSC 13 MC

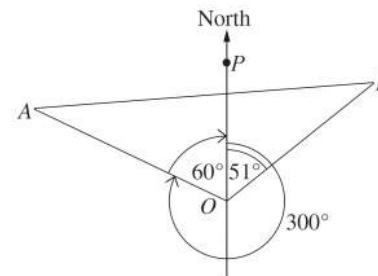


$$\begin{aligned}\text{Bearing of A from B} &= 180 + 120 \\ &= 300^\circ \\ &\Rightarrow D\end{aligned}$$

♦ Mean mark 49%

Worked Solutions

3. Measurement, STD2 M6 2015 HSC 7 MC



Let P be due North of O

$$\angle POB = 51^\circ$$

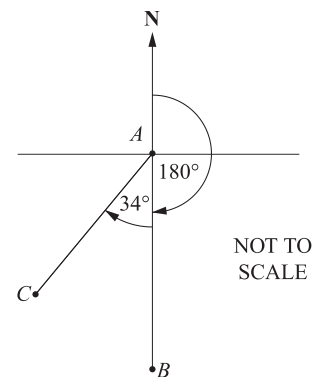
$$\begin{aligned}\angle POA &= 360 - 300 \\ &= 60^\circ\end{aligned}$$

$$\begin{aligned}\therefore \angle AOB &= 51 + 60 \\ &= 111^\circ\end{aligned}$$

$$\Rightarrow B$$

4. Measurement, STD2 M6 2018 HSC 7 MC

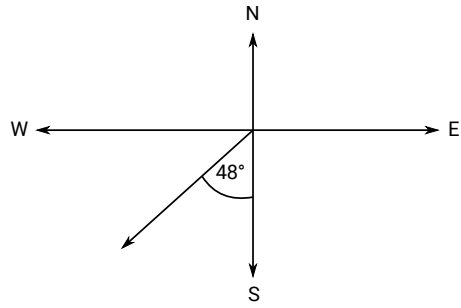
Bearing of Town C from Town A :



$$\begin{aligned}\text{Bearing} &= 180 + 34 \\ &= 214^\circ\end{aligned}$$

$$\Rightarrow C$$

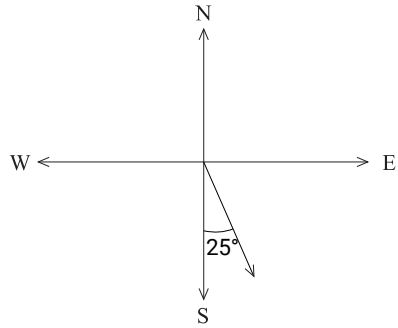
5. Measurement, STD2 M6 2022 HSC 8 MC



$$\begin{aligned}\text{True bearing} &= 180 + 48 \\ &= 228^\circ\end{aligned}$$

$\Rightarrow C$

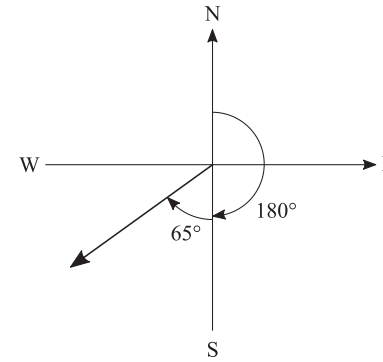
6. Measurement, STD2 M6 SM-Bank 2 MC



$$\begin{aligned}S60^\circ W &= 180 - 25 \\ &= 155^\circ\end{aligned}$$

$\Rightarrow D$

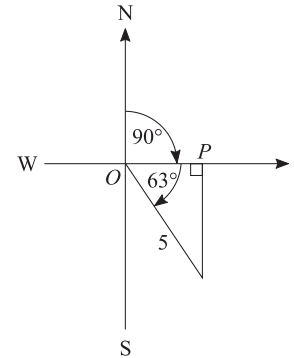
7. Measurement, STD2 M6 SM-Bank 3 MC



$$\begin{aligned}\text{True bearing} &= 180 + 65 \\ &= 245^\circ\end{aligned}$$

$\Rightarrow C$

8. Measurement, STD2 M6 SM-Bank 7 MC



Jeet finishes at P

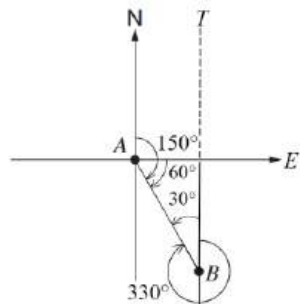
Find OP :

$$\cos 63^\circ = \frac{OP}{5}$$

$$\begin{aligned}\therefore OP &= 5 \times \cos 63^\circ \\ &= 2.26 \dots\end{aligned}$$

$\Rightarrow A$

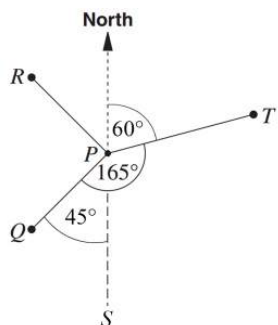
9. Measurement, STD2 M6 2010 HSC 10 MC



♦♦ Mean mark 34%

$\angle TBA = 30^\circ$ (angle sum of triangle)
 $\therefore$ Bearing of A from B
 $= 360 - 30$
 $= 330^\circ$
 $\Rightarrow D$

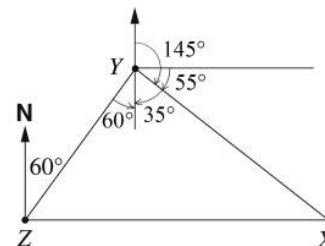
10. Measurement, STD2 M6 2008 HSC 17 MC



$\angle QPS = 45^\circ$ (Q is south west of P)
 $\angle TPS = 165 - 45 = 120^\circ$
 $\therefore \angle NPT = 60^\circ$ (180° in straight line)
 $\Rightarrow A$

11. Measurement, STD2 M6 2014 HSC 23 MC

Since X is due east of Z
 $\Rightarrow$ Cannot be B or D



♦ Mean mark 38%
COMMENT: Drawing a parallel North/South line through Y makes this question **much simpler** to solve.

The diagram shows we can find
 $\angle ZYX = 60 + 35^\circ = 95^\circ$
 Using alternate angles (60°) and
 the 145° bearing of X from Y
 $\Rightarrow C$

12. Measurement, STD2 M6 2016 HSC 25 MC

Using the sine rule,

$$\frac{\sin \angle BCA}{40} = \frac{\sin 25^\circ}{18}$$

$$\sin \angle BCA = \frac{40 \times \sin 25^\circ}{18}$$

$$= 0.939 \dots$$

$$\angle BCA = 180 - 69.9 \quad (\angle BCA > 90^\circ)$$

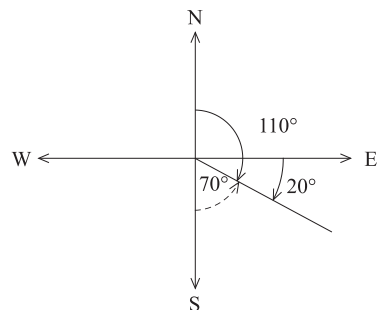
$$= 110.1^\circ$$

$\therefore$ Bearing of C from B
 $= 110.1 + 25$ (external angle of triangle)
 $= 135.1$

$\Rightarrow D$

♦ Mean mark 39%.

13. Measurement, STD2 M6 2019 HSC 4 MC

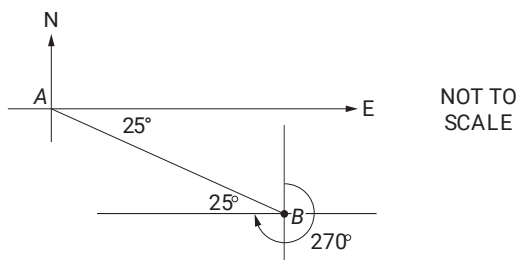


♦ Mean mark 47%.

$$110^\circ = S70^\circ E$$

$$\Rightarrow C$$

14. Measurement, STD2 M6 2021 HSC 14 MC



♦♦ Mean mark 28%.

$$\text{Bearing (A from B)} = 270 + 25$$

$$= 295^\circ$$

$\Rightarrow D$

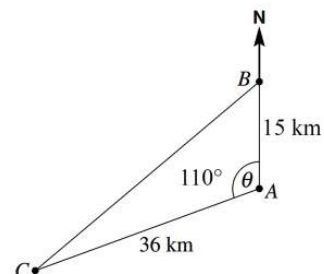
15. Measurement, STD2 M6 2005 HSC 27c

i. There is 360° about point A

$$\therefore \theta + 250^\circ = 360^\circ$$

$$\theta = 110^\circ$$

ii.



$$a^2 = b^2 + c^2 - 2ab \cos A$$

$$CB^2 = 36^2 + 15^2 - 2 \times 36 \times 15 \times \cos 110^\circ$$

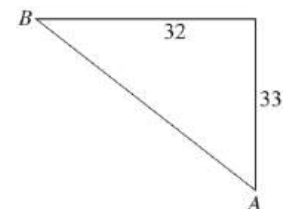
$$= 1296 + 225 - (-369.38\dots)$$

$$= 1890.38\dots$$

$$\therefore CB = 43.47\dots$$

$$= 43 \text{ km (nearest km)}$$

16. Measurement, STD2 M6 2016 HSC 26d



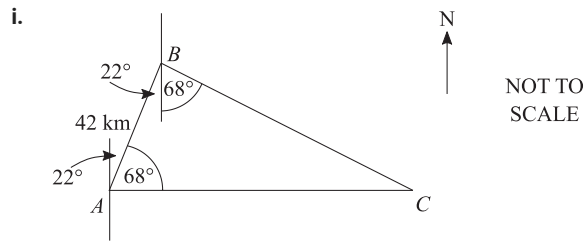
Using Pythagoras,

$$AB = \sqrt{32^2 + 33^2}$$

$$= 45.967\dots$$

$$= 46 \text{ m (nearest m)}$$

17. Measurement, STD2 M6 SM-Bank 4



$$\begin{aligned}\angle ABC &= 22 + 68 \\ &= 90^\circ\end{aligned}$$

ii. In $\triangle ABC$,

$$\begin{aligned}\cos \angle BAC &= \frac{AB}{AC} \\ \cos 68^\circ &= \frac{42}{AC} \\ AC &= \frac{42}{\cos 68^\circ} \\ &= 112.11\dots \\ &= 112 \text{ km (nearest km)}\end{aligned}$$

iii. Travel time = $\frac{\text{dist}}{\text{speed}}$

$$\begin{aligned}&= \frac{42}{12.6} \\ &= 3.333\dots \\ &= 3\text{h } 20 \text{ mins}\end{aligned}$$

18. Measurement, STD2 M6 2020 HSC 31

a. $\angle APB = 100 - 35$
 $= 65^\circ$

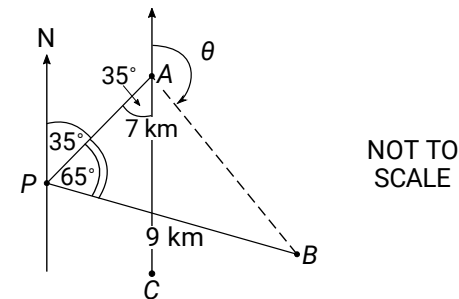
b. Using cosine rule:

$$\begin{aligned}AB^2 &= AP^2 + PB^2 - 2 \times AP \times PB \cos 65^\circ \\ &= 49 + 81 - 2 \times 7 \times 9 \cos 65^\circ \\ &= 76.750\dots\end{aligned}$$

$$\begin{aligned}\therefore AB &= 8.760\dots \\ &= 8.76 \text{ km (to 2 d.p.)}\end{aligned}$$

Mean mark 53%.

c.



$$\angle PAC = 35^\circ \text{ (alternate)}$$

Using cosine rule, find $\angle PAB$:

$$\begin{aligned}\cos \angle PAB &= \frac{7^2 + 8.76 - 9^2}{2 \times 7 \times 8.76} \\ &= 0.3647\dots\end{aligned}$$

$$\begin{aligned}\therefore \angle PAB &= 68.61\dots^\circ \\ &= 69^\circ \text{ (nearest degree)}\end{aligned}$$

$$\begin{aligned}\therefore \text{Bearing of } B \text{ from } A (\theta) &= 180 - (69 - 35) \\ &= 146^\circ\end{aligned}$$

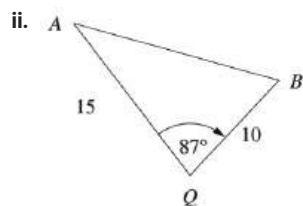
♦♦ Mean mark 22%.

19. Measurement, STD2 M6 2007 HSC 26a

i. 2 hrs 48 mins = 168 mins

$$\begin{aligned}\text{Speed (A to Q)} &= \frac{15}{168} \\ &= 0.0892\dots \text{ km/min}\end{aligned}$$

$$\begin{aligned}\text{Speed (in km/hr)} &= 0.0892\dots \times 60 \\ &= 5.357\dots \text{ km/hr} \\ &= 5 \text{ km/hr (nearest km/hr)}\end{aligned}$$

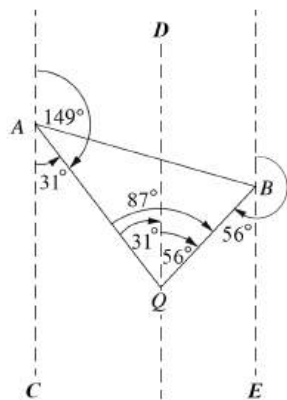


Using cosine rule

$$\begin{aligned}AB^2 &= 15^2 + 10^2 - 2 \times 15 \times 10 \times \cos 87^\circ \\ &= 309.299\dots \\ AB &= 17.586\dots \\ &= 18 \text{ km (nearest km)}\end{aligned}$$

$\therefore$ The distance from Town A to Town B is 18 km.

iii



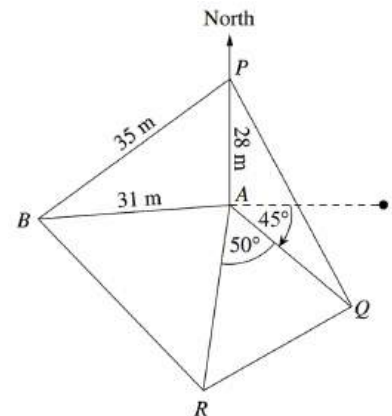
$$\begin{aligned}\angle CAQ &= 31^\circ \text{ (straight angle at A)} \\ \angle AQD &= 31^\circ \text{ (alternate angle } AC \parallel DQ) \\ \angle DQB &= 87 - 31 = 56^\circ\end{aligned}$$

$$\angle QBE = 56^\circ \text{ (alternate angle } DQ \parallel BE)$$

$$\begin{aligned}\therefore \text{Bearing of Q from B} &= 180 + 56 \\ &= 236^\circ\end{aligned}$$

20. Measurement, STD2 M6 2004 HSC 24b

i.



Let E be directly east of A

$$\angle EAQ = 45^\circ \text{ (given)}$$

$$\begin{aligned}\therefore \angle PAQ &= 90^\circ + 45^\circ \\ &= 135^\circ\end{aligned}$$

ii. Bearing of R from A

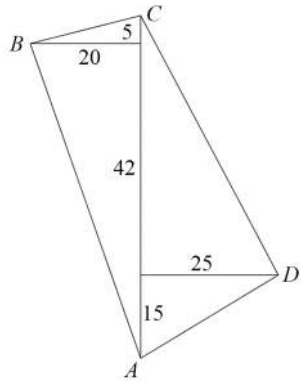
$$\begin{aligned}&= 90^\circ + 45^\circ + 50^\circ \\ &= 185^\circ \text{ (or S } 5^\circ \text{ W)}\end{aligned}$$

iii. Using cosine rule in $\triangle PAB$

$$\begin{aligned}\cos \angle PAB &= \frac{31^2 + 28^2 - 35^2}{2 \times 31 \times 28} \\ &= 0.2995\dots \\ \therefore \angle PAB &= 72.570\dots^\circ \\ &= 73^\circ \text{ (nearest degree)}\end{aligned}$$

21. Measurement, STD2 M6 2006 HSC 26a

i.



ii. Using Pythagoras:

$$CD^2 = 47^2 + 25^2$$

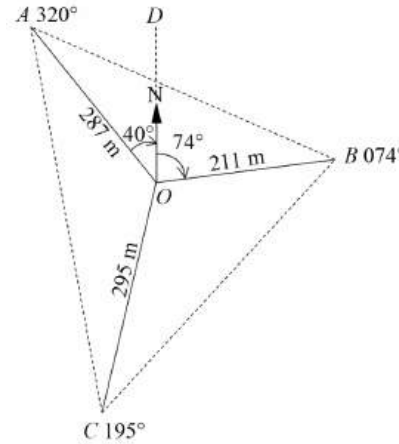
$$= 2834$$

$$\therefore CD = 53.235\dots$$

$$= 53 \text{ m (nearest m)}$$

22. Measurement, STD2 M6 2014 HSC 28b

i.



Let D be directly north of O

$$\angle AOD = 360 - 320 = 40^\circ$$

$$\therefore \angle AOB = 40 + 74 = 114^\circ \dots \text{as required}$$

ii. Using cosine rule

$$AB^2 = AO^2 + BO^2 - 2 \times AO \times BO \times \cos \angle AOB$$

$$= 287^2 + 211^2 - 2 \times 287 \times 211 \times \cos 114^\circ$$

$$= 126\,890 - 121\,114(-0.4067\dots)$$

$$= 176\,151.50\dots$$

$$AB = 419.704\dots$$

$$= 420 \text{ m (nearest m)}$$

iii. Using $A = \frac{1}{2}ab\sin C$,

$$\text{Area } \triangle AOB = \frac{1}{2} \times 287 \times 211 \times \sin 114^\circ$$

$$= 27\,660.786\dots \text{ m}^2$$

$$= 2.7660\dots \text{ ha} \quad (1 \text{ ha} = 10\,000 \text{ m}^2)$$

$$= 2.8 \text{ ha (2 sig figures)}$$

♦ Mean marks of 42% and 41% for parts (ii) and (iii) respectively.

23. Measurement, STD2 M6 2023 HSC 27

a. In $\triangle XPY$:

$$\angle PXY = 180 - 120 = 60^\circ$$

$$\cos 60^\circ = \frac{XP}{15}$$

$$\begin{aligned} XP &= 15 \times \cos 60^\circ \\ &= 7.5 \text{ km} \end{aligned}$$

b. In $\triangle XPC$:

$$\text{Let } \theta = \angle CXP$$

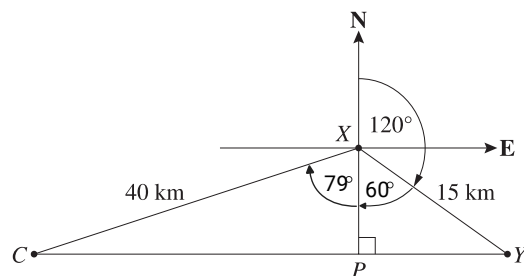
$$\cos \theta = \frac{7.5}{40}$$

$$\theta = \cos^{-1} \left(\frac{7.5}{40} \right)$$

$$= 79.193 \dots$$

$$= 79^\circ \text{ (nearest degree)}$$

♦ Mean mark (b) 39%.



$$\begin{aligned} \text{Bearing } C \text{ from } X &= 180 + 79 \\ &= 259^\circ \end{aligned}$$

24. Measurement, STD2 M6 2013 HSC 28a

$$\begin{aligned} \angle AOB &= 21^\circ + (360 - 310) \\ &= 71^\circ \end{aligned}$$

$$\text{Using Area} = \frac{1}{2} ab \sin C$$

$$\begin{aligned} \text{Area} &= \frac{1}{2} \times 60 \times 75 \times \sin 71^\circ \\ &= 2127.4167 \dots \\ &= 2127 \text{ m}^2 \text{ (nearest m}^2\text{)} \end{aligned}$$

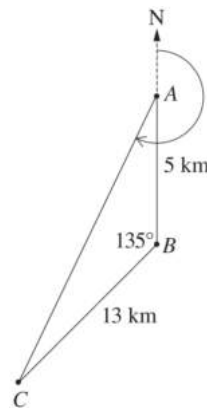
♦ Mean mark 50%

25. Measurement, STD2 M6 2017 HSC 30c

i. Using cosine rule:

$$\begin{aligned} AC^2 &= AB^2 + BC^2 - 2 \times AB \times BC \times \cos 135^\circ \\ &= 5^2 + 13^2 - 2 \times 5 \times 13 \times \cos 135^\circ \\ &= 285.923 \dots \\ \therefore AC &= 16.909 \dots \\ &= 17 \text{ km (nearest km)} \end{aligned}$$

ii.



Using sine rule, find $\angle BAC$:

$$\begin{aligned} \frac{\sin \angle BAC}{13} &= \frac{\sin 135^\circ}{17} \\ \sin \angle BAC &= \frac{13 \times \sin 135^\circ}{17} \\ &= 0.5407 \dots \\ \angle BAC &= 32.7^\circ \end{aligned}$$

$$\begin{aligned} \therefore \text{Bearing of } C \text{ from } A &= 180 + 32.7 \\ &= 212.7^\circ \\ &= 213^\circ \end{aligned}$$

♦♦ Mean mark 31%.

26. Measurement, STD2 M6 2021 HSC 39

a. Using sine rule:

$$\text{Area} = \frac{1}{2}ab\sin C$$

$$466 = \frac{1}{2} \times 28 \times 35 \times \sin \angle COB$$

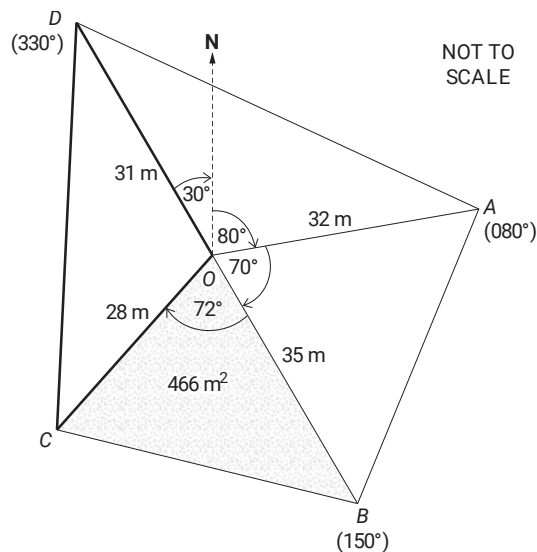
$$\sin \angle COB = \frac{466}{490}$$

$$\angle COB = \sin^{-1} \left(\frac{466}{490} \right)$$

$$= 71.99$$

$$= 72^\circ \text{ (nearest degree)}$$

b.



$$\begin{aligned} \angle DOC &= 360 - (30 + 80 + 70 + 72) \\ &= 108^\circ \end{aligned}$$

Using the cosine rule:

$$\begin{aligned} DC^2 &= 31^2 + 28^2 - 2 \times 31 \times 28 \times \cos 108^\circ \\ &= 2281.4535 \end{aligned}$$

$$DC = 47.76 \dots \text{ m}$$

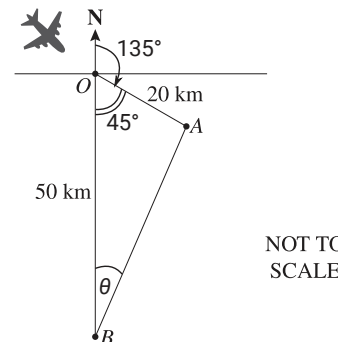
$$\begin{aligned} \therefore \text{Length of fence} &= 47.8 + 31 + 28 \\ &= 106.8 \text{ m (1 d.p.)} \end{aligned}$$

♦ Mean mark part (a) 34%.

♦♦ Mean mark part (b) 33%.

27. Measurement, STD2 M6 2022 HSC 33

a. $\angle AOB = 180 - 135 = 45^\circ$



Using the cosine rule:

$$\begin{aligned} AB^2 &= 20^2 + 50^2 - 2 \times 20 \times 50 \times \cos 45^\circ \\ &= 1485.786 \dots \end{aligned}$$

$$\begin{aligned} \therefore AB &= 38.54 \dots \\ &= 38.5 \text{ km (to 1 d.p.)} \end{aligned}$$

b. Let $\theta = \angle OBA$

Using the sine rule:

$$\frac{\sin \theta}{20} = \frac{\sin 45^\circ}{38.5}$$

$$\sin \theta = \frac{\sin 45^\circ \times 20}{38.5}$$

$$\therefore \theta = \sin^{-1} \left(\frac{\sin 45^\circ \times 20}{38.5} \right)$$

$$= 21.55 \dots^\circ$$

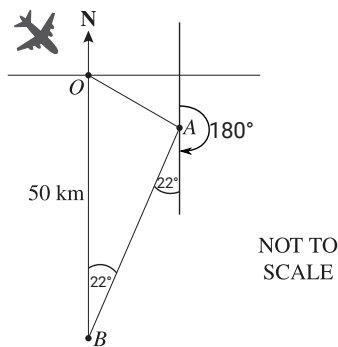
$$= 22^\circ \text{ (nearest degree)}$$

c.

♦ Mean mark part (a) 50%.

♦♦ Mean mark part (b) 39%.

♦♦ Mean mark part (c) 25%.



$$\begin{aligned}\text{Bearing of } B \text{ from } A \\ &= 180 + 22 \\ &= 202^\circ \text{T}\end{aligned}$$

28. Measurement, STD2 M6 2019 HSC 35

In $\triangle COB$,

$$\text{Area} = \frac{1}{2}absinc$$

$$198 = \frac{1}{2} \times 16 \times 25 \times \sin \angle COB$$

$$\begin{aligned}\sin \angle COB &= \frac{2 \times 198}{16 \times 25} \\ &= 0.99\end{aligned}$$

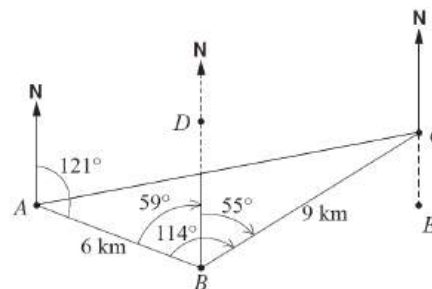
$$\begin{aligned}\angle COB &= \sin^{-1} 0.99 \\ &= 81^\circ 53'\end{aligned}$$

$$\begin{aligned}\therefore \text{Bearing of } C \text{ from } O &= 125^\circ + 81^\circ 53' \\ &= 206^\circ 53' \\ &= 207^\circ \text{ (nearest degree)}\end{aligned}$$

♦♦ Mean mark 26%.

29. Measurement, STD2 M6 2011 HSC 24c

i.



Let point D be due North of point B

$$\angle ABD = 180 - 121 \text{ (cointerior with } \angle A) = 59^\circ$$

$$\angle DBC = 114 - 59 = 55^\circ$$

$\therefore$ Bearing of C from B is 055°

ii. Using cosine rule:

$$\begin{aligned}AC^2 &= AB^2 + BC^2 - 2 \times AB \times BC \times \cos \angle ABC \\ &= 6^2 + 9^2 - 2 \times 6 \times 9 \times \cos 114^\circ \\ &= 160.9275 \dots\end{aligned}$$

$$\begin{aligned}\therefore AC &= 12.685 \dots \text{ (Noting } AC > 0) \\ &= 13 \text{ km (nearest km)}\end{aligned}$$

iii. Need to find $\angle ACB$ (see diagram)

$$\begin{aligned}\cos \angle ACB &= \frac{AC^2 + BC^2 - AB^2}{2 \times AC \times BC} \\ &= \frac{(12.685 \dots)^2 + 9^2 - 6^2}{2 \times (12.685 \dots) \times 9} \\ &= 0.9018 \dots \\ \angle ACB &= 25.6^\circ \text{ (to 1 d.p.)}\end{aligned}$$

From diagram,

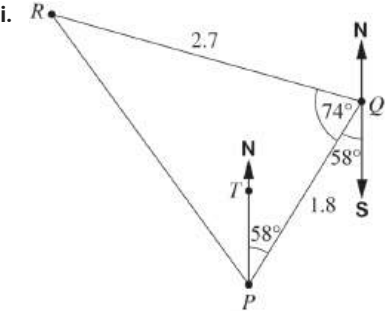
$$\angle BCE = 55^\circ \text{ (alternate angle, } DB \parallel CE)$$

$$\begin{aligned}\therefore \text{Bearing of } A \text{ from } C \\ &= 180 + 55 + 25.6 \\ &= 260.6 \\ &= 261^\circ \text{ (nearest degree)}\end{aligned}$$

STRATEGY: Important: Draw North-South parallel lines through major points to make the angle calculations easier!

MARKER'S COMMENT: The best responses clearly showed what steps were taken with working on the diagram. Note that all North/South lines are parallel.

30. Measurement, STD2 M6 2009 HSC 27b



$\angle PQS = 58^\circ$ (alternate to $\angle TPQ$)
Bearing of R from Q
 $= 180^\circ + 58^\circ + 74^\circ$
 $= 312^\circ$

ii. Using Cosine rule:

$$RP^2 = RQ^2 + PQ^2 - 2 \times RQ \times PQ \times \cos \angle RQP$$
$$= 2.7^2 + 1.8^2 - 2 \times 2.7 \times 1.8 \times \cos 74^\circ$$
$$= 7.29 + 3.24 - 2.679\dots$$
$$= 7.851\dots$$
$$\therefore RP = \sqrt{7.851\dots}$$
$$= 2.8019\dots$$
$$\approx 2.8 \text{ km (1 d.p.)}$$

iii. Using $A = \frac{1}{2}ab\sin C$

$$A = \frac{1}{2} \times 2.7 \times 1.8 \times \sin 74^\circ$$
$$= 2.3358\dots$$
$$= 2.3 \text{ km}^2$$

$$\therefore \text{No-go zone is } 2.3 \text{ km}^2$$

◆◆◆ Mean mark 18%.
TIP: Draw North-South parallel lines through relevant points to help calculate angles as shown in the Worked Solutions.

◆ Mean mark 36%

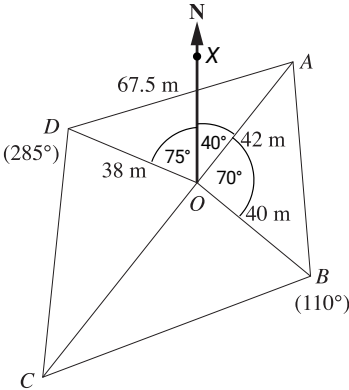
◆ Mean mark 44%

31. Measurement, STD2 M6 2024 HSC 40

In $\triangle ODA$, using cosine rule:

$$\cos \angle DOA = \frac{38^2 + 42^2 - 67.6^2}{2 \times 38 \times 42}$$
$$= -0.4223\dots$$
$$\angle DOA = 115.0^\circ$$

Let X be a point directly north of point O :



$$\angle DOX = 360 - 285 = 75^\circ$$
$$\angle XOA = 115 - 75 = 40^\circ$$
$$\angle AOB = 110 - 40 = 70^\circ$$
$$\angle COB = 180 - 70 = 110^\circ \text{ (AOC is a straight line)}$$
$$\therefore \text{Bearing of } C \text{ from } O = 110 + 110 = 220^\circ$$

STRATEGY: The information in $\triangle ODA$ strongly suggests finding $\angle DOA$ is a good strategy to explore.